

OriginalAuto — AI Developer Master Project Specification

Version 1.0 — Master Blueprint & Implementation Governance Document

Production domain: <https://originalauto.ir> — Previous/internal name: Original Shargh

This document is an architectural specification. It is NOT permission to modify the project.

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1. Purpose, Reading Rules & Data Classification

This document is the authoritative blueprint for **OriginalAuto**, a Persian-language automotive repair and technical knowledge platform for the Iranian automotive market. It will be handed to an AI software-development agent together with the actual project repository. The document defines the intended architecture precisely enough that the agent must not need to guess — and it explicitly separates what is confirmed from what is proposed, future, or unverified.

The most important rule in this document: the repository is the ground truth for what exists today. This document is the ground truth for what is intended. Where the two differ, the AI developer reports the difference in the audit phases — it never silently "fixes" either side.

1.1 Data Classification Legend

Every substantive statement in this document carries one of the following tags. The AI developer must respect them mechanically.

Tag	Meaning	Required treatment
[CONFIRMED]	Fact stated by the project owner about the current project or firm intent	Verify against the actual repository before writing code or SQL; repository reality wins on "what exists"
[PROPOSED]	Recommended architecture, not yet verified to exist in the repo	Adopt only after audit phases and explicit approval
[FUTURE]	Roadmap capability, out of scope for early phases	Do not implement now; design so it can be added without rework
[NEEDS VERIFICATION]	Data that must be checked against authoritative sources or the real schema before use	Never publish, insert, or treat as authoritative before verification
[DO NOT INSERT]	Data forbidden in production without a verified technical source	Hard block — see per-section rules

READ FIRST. AUDIT SECOND. PLAN THIRD. ASK FOR APPROVAL FOR HIGH-RISK CHANGES. IMPLEMENT FOURTH. TEST FIFTH. REPORT LAST.

Critical Workflow Rule

No code may be modified before audit Phases A–I are complete and reviewed. Every major change must be minimal, reversible, documented, and testable.

2. Project Overview

OriginalAuto is a **[CONFIRMED]** custom PHP/MySQL MVC application — **not Laravel** — operating in production at <https://originalauto.ir>. Its previous/internal name is **Original Shargh [CONFIRMED]**. It is a Persian-language automotive repair and technical knowledge platform.

The long-term vision is a large automotive repair knowledge base structured like a specialized automotive encyclopedia. The platform is **Vehicle-First**: the user starts from the vehicle and navigates outward.

2.1 Core User Journey **[CONFIRMED intent]**

Brand → Model → Vehicle details → Services → Problems → Symptoms → DTC → Repair Solutions → Parts → Articles → FAQ

2.2 Platform Priorities **[CONFIRMED]**

- Technical accuracy
- Original, useful content
- SEO
- Internal linking
- Structured automotive data
- Real relationships between vehicles and services
- Scalability
- Maintainability
- Persian language support
- Mobile usability
- Production safety

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Master service categories
defined

45

Candidate brand groups
(NEEDS VERIFICATION)

~190

Candidate model names
(NEEDS VERIFICATION)

15

Diagnostic subservices
defined

2.3 Public Release Status **[CONFIRMED]**

For the current public release, the following sections are temporarily hidden:

1. Store / Shop
2. Repair Shop Dashboard
3. User Account

These sections must not accidentally appear in navigation, and must be handled correctly from an SEO/indexability perspective. **[PROPOSED]** handling: keep them out of navigation, exclude from sitemap, and serve `noindex` until launch — verify current implementation during Phase G.

2.4 The Goal

The goal is not merely to create many pages. The goal is to create a technically useful, scalable, SEO-friendly automotive knowledge platform for the Iranian market.

3. Critical Development Workflow & Audit Phases

3.1 The Do-Not-Touch-Yet Rule **[CONFIRMED]**

When the repository is provided, the AI developer must **not** modify anything immediately. It must first inspect:

1. The entire repository
2. The MVC architecture
3. Routes
4. Controllers
5. Models
6. Views
7. Configuration
8. Database schema
9. Generated data
10. SEO implementation
11. JavaScript and frontend URLs
12. .htaccess / Apache rules
13. Production configuration

14. Current vehicle architecture

15. Current service architecture

Only after inspection does it produce the audit deliverables:

Deliverable	Scope
A. Repository Audit	Structure, MVC, routing, config, frontend
B. Database Audit	Real schema, columns, relationships, data
C. Vehicle Architecture Audit	Brand/model/hierarchy reality vs. this spec
D. Service Architecture Audit	Services and taxonomy reality
E. URL Architecture Audit	Live URL system vs. target structure
F. SEO Audit	Metadata, canonicals, sitemap, robots, schema
G. Content Architecture Audit	Content model, generated data, thin content
H. Security / Production Risk Audit	Credentials, config, exposure, safety
I. Architecture Gap Report	Section 25 template, filled from findings

Only after that may implementation planning (Phase J) begin.

3.2 Implementation Phases [CONFIRMED]

Phase	Name	Gate to pass
A	Repository Audit	All 15 inspection points done
B	Database Audit	Real schema documented
C	Vehicle Catalog Audit	Hierarchy mapped to real tables
D	Service Taxonomy Audit	Service model documented
E	Vehicle-Service Matrix Audit	Canonical relationship verified
F	Routing Audit	URL system documented
G	SEO Audit	SEO behavior documented
H	Content Architecture Audit	Content model documented
I	Gap Report	Section 25 table filled, reviewed
J	Implementation Plan	Approved before any change

Phase	Name	Gate to pass
K	Low-Risk Implementation	Only approved, minimal changes
L	Testing	Section 23 test suite passes
M	Production Verification	Section 26 final report delivered

Do not skip the audit phases. Do not begin Phase K before Phases A–I are reviewed and Phase J is approved.

4. Confirmed Current Architecture

4.1 Stack [CONFIRMED]

The project is a custom PHP/MySQL MVC application. The following components were stated by the project owner:

Component	Path / Concept	Role
Front controller	<code>index.php</code>	Single entry point
Router	<code>app/Core/Router.php</code>	Request dispatch
Database layer	<code>app/Core/Database.php</code>	DB connection/queries
Base controller	<code>app/Core/Controller.php</code>	Shared controller logic
Web routes	<code>app/routes.php</code>	Route definitions
API routes	<code>app/api_routes.php</code>	API route definitions
Application layers	controllers, models, views	Standard MVC
Configuration	<code>config</code>	App configuration
Generated data	<code>app/Data/generated_models</code> , <code>app/Data/generated_matrices</code>	Pre-generated content/data files

[NEEDS VERIFICATION] Exact file inventory, class responsibilities, route tables, and config contents must be confirmed in Phase A. The table above is the stated structure, not a verified file listing.

4.2 Framework Rules [CONFIRMED]

- The project is **NOT Laravel**. Do not introduce Laravel architecture.

- Do not create unnecessary frameworks or rewrite the application unless the audit proves it is necessary.
- Do not create duplicate architecture.
- Do not create a second competing vehicle-service relationship system.
- Do not remove existing working functionality without analysis.

5. Database Architecture & Canonical Relationships

5.1 Known Database Concepts [CONFIRMED concepts – real schema NEEDS VERIFICATION]

Table concept	Role
<code>vehicle_brands</code>	Brands
<code>vehicle_models</code>	Models
<code>vehicle_model_service</code>	Canonical vehicle ↔ service relationship
<code>services</code>	Service catalog
<code>vehicle_common_problems</code>	Per-model common problems
<code>vehicle_dtc_codes</code>	DTC data tied to vehicles
<code>vehicle_diagnostics</code>	Diagnostic data
<code>vehicle_model_faqs</code>	Per-model FAQ
<code>vehicle_maintenance_tasks</code>	Maintenance tasks
<code>vehicle_ecu_info</code>	ECU/module information
<code>vehicle_repair_solutions</code>	Repair solutions
<code>vehicle_symptoms</code>	Symptoms
<code>articles</code>	Articles
<code>diagnostic_knowledge</code>	Diagnostic knowledge base
related tables	To be discovered in Phase B

5.2 Canonical Relationship Rules [CONFIRMED]

1. `vehicle_model_service` is the **single source of truth** for the relationship between a vehicle model and a service. It must remain so.

2. Do **NOT** create a new table such as `service_vehicle_matrix` if `vehicle_model_service` already provides the required relationship.
3. Generated files (Section 19) may enrich content, but must not replace the canonical database relationship.
4. The AI developer must inspect the real schema **before writing any SQL**.

5.3 Database Safety Rules [CONFIRMED – hard blocks]

- DO NOT `DROP` tables.
- DO NOT `TRUNCATE` tables.
- DO NOT delete production data.
- DO NOT perform destructive migrations.
- DO NOT modify the production database schema without approval.
- Any schema change requires the mapping protocol in Section 22 first, including a rollback strategy.

6. Vehicle Hierarchy (Target Model)

6.1 Conceptual Hierarchy [PROPOSED target]

Manufacturer / Brand → Market Brand → Model → Generation → Variant / Trim → Engine → Transmission → ECU / Modules → Services → Problems → Symptoms → DTC → Repair Solutions → Parts → Articles → FAQ

6.2 Mapping Conceptual Hierarchy to the Real Schema

[CONFIRMED rule] Do not assume all of these entities already exist as database tables. The AI developer must map the conceptual architecture onto the existing schema first. If new schema is required, report it before implementation. Do not create new tables automatically.

Level	Conceptual entity	Candidate existing table	Status
1	Manufacturer / Brand	?	NEEDS VERIFICATION
2	Market Brand	<code>vehicle_brands</code> ?	NEEDS VERIFICATION – confirm whether it stores market brands, manufacturers, or both
3	Model	<code>vehicle_models</code>	CONFIRMED concept
4	Generation	none known	FUTURE – schema decision required
5	Variant / Trim	none known	FUTURE

Level	Conceptual entity	Candidate existing table	Status
6	Engine	none known	FUTURE
7	Transmission	none known	FUTURE
8	ECU / Modules	vehicle_ecu_info	CONFIRMED concept
9	Services	services + vehicle_model_service	CONFIRMED
10	Problems	vehicle_common_problems	CONFIRMED concept
11	Symptoms	vehicle_symptoms	CONFIRMED concept
12	DTC	vehicle_dtc_codes	CONFIRMED concept
13	Repair Solutions	vehicle_repair_solutions	CONFIRMED concept
14	Parts	none known	FUTURE
15	Articles	articles	CONFIRMED concept
16	FAQ	vehicle_model_faqs	CONFIRMED concept
–	Maintenance	vehicle_maintenance_tasks	CONFIRMED concept
–	Diagnostics	vehicle_diagnostics , diagnostic_knowledge	CONFIRMED concepts

7. Brand & Model Catalog

7.1 Catalog Rules [CONFIRMED]

- The catalog should eventually cover the Iranian automotive market comprehensively.
- It must account for: historical vehicles, discontinued vehicles, imported vehicles, limited-import vehicles, locally assembled vehicles, rebadged vehicles, Chinese brands, and Iranian market badges.
- Do **not** blindly treat an assembler, a manufacturer, and a market badge as identical.
- Do **not** claim the catalog is complete. Other brands/models must be researched and verified before being added to the final master catalog.

7.2 Manufacturer vs. Market Brand [NEEDS VERIFICATION]

A vehicle may simultaneously have a Manufacturer (e.g., Chinese), an Iranian market brand/badge, an Iranian importer/assembler, and an actual model. The database architecture must not collapse these concepts.

Market brand	Stated possible source/manufacturer	Status
Fownix	Chery	NEEDS VERIFICATION
KMC	JAC or other source manufacturers	NEEDS VERIFICATION
Lamari	Dongfeng / Forthing	NEEDS VERIFICATION
Xtrim	Exeed	NEEDS VERIFICATION
Fidelity	Jetour	NEEDS VERIFICATION
Lucano	Jaecoo	NEEDS VERIFICATION

These are the project owner's stated hypotheses. **[DO NOT INSERT]** They must be verified before being treated as canonical data.

7.3 Candidate Brand Groups [NEEDS VERIFICATION – list not exhaustive]

Iran Khodro, Saipa, Pars Khodro, Zamyad, Bahman, Kerman Motor, MVM, Chery, Fownix, Xtrim, KMC, JAC, Haima, FMC, Lamari, Changan, Dongfeng, BYD, Geely, GAC, Haval, Foton, MG, Toyota, Lexus, Nissan, Mitsubishi, Honda, Suzuki, Hyundai, Kia, Renault, Volkswagen, BMW, Mercedes-Benz, Audi, Volvo, Land Rover, Porsche, Mini, Jaguar, Genesis, Jeep, Chevrolet, Cadillac – plus newer/imported brands present in Iran.

7.4 Example Vehicle Catalog [NEEDS VERIFICATION – candidate dataset, not canonical]

Group	Candidate models (owner-provided)
IKCO	Peugeot 405, Pars, 206, 206 SD, 207, 2008, Samand, Samand LX, Soren, Soren Plus, Dena, Dena Plus, Runna, Runna Plus, Tara (V1, V4), Reera, Arisan, Arisan 2
SAIPA	Pride (111, 131, 132, 141, 151), Tiba, Tiba 2, Saina, Saina S, Quick, Quick R, Quick S, Shahin, Shahin G, Atlas, Atlas G, Sahand
Pars Khodro	Tondar 90, Tondar Plus, Sandero, Sandero Stepway, Megane; Brilliance H220/H230/H320/H330/C3
JAC / KMC	J4, J5, S3, S5, J7, X5, JS6, K7, T8, T9, EJ7

Group	Candidate models (owner-provided)
MVM / Chery / Fownix	MVM 110, 110S, 315, 315 Plus, 530, 550; X22, X22 Pro, X33, X33 Cross, X55, X55 Pro; Tiggo 5/7/8; Arrizo 5/6/8; FX
FMC	SX5, SX5 Plus, T5, M4, 511
Lamari	EAMA, EAMA Hybrid
Bahman	Mazda 323, Mazda 3, Mazda 3 New; B30, B50; Capra, Capra 2; Dignity, Dignity Prestige; Fidelity, Fidelity Prestige; Respect, Respect 2
Toyota	Corolla, Camry, Yaris, Prius, C-HR, RAV4, Highlander, Land Cruiser, Prado, Hilux, FJ Cruiser
Lexus	IS, ES, GS, LS, NX, RX, LX, CT
Hyundai	i10, i20, Accent, Elantra, Sonata, Azera, Tucson, ix35, Santa Fe, Veracruz, Kona, Palisade
Kia	Pride, Rio, Cerato, Forte, Optima, K5, Sportage, Sorento, Mohave, Soul, Picanto, Carnival
Nissan	Patrol, Maxima, Teana, Murano, X-Trail, Qashqai, Juke, Sunny, Pickup
Renault	5, 21, Logan, Sandero, Stepway, Megane, Fluence, Latitude, Koleos, Captur, Duster
Volkswagen	Golf, Passat, Jetta, Tiguan, Touareg, Beetle, T-Roc
BMW	1–7 Series; X1, X3, X4, X5, X6, X7
Mercedes-Benz	A-Class, B-Class, C-Class, E-Class, S-Class, CLA, GLA, GLC, GLE, GLS, G-Class

Disambiguation warning: the same model name appears under multiple groups in this list (e.g., Pride under both SAIPA and Kia; Sandero/Stepway/Megane under both Pars Khodro and Renault; Tondar/Megane being market-badged Renault models). This is exactly the market-badge problem of Section 7.2 and must be resolved per record during the master data workflow. **[DO NOT INSERT]** unverified records.

8. Master Service Taxonomy

8.1 Top-Level Service Categories [CONFIRMED intent]

#	Service	Scope
1	Diagnostic	OBD scan, DTC reading/analysis, live data, module-level diagnosis
2	ECU Programming	Flashing, coding, adaptation, immobilizer/key work
3	Engine Repair	Mechanical, timing, head, overhaul

#	Service	Scope
4	Periodic Service	Scheduled maintenance
5	Electrical	Battery, charging, starting, wiring, modules
6	Gearbox	Manual, automatic, CVT, DCT, AMT
7	Fuel System	Pump, injectors, sensors, EVAP
8	Cooling System	Radiator, pump, thermostat, fans
9	Brakes	Pads, discs, hydraulics, ABS/EBD/ESP
10	Suspension	Dampers, arms, bushings
11	Steering	Hydraulic/EPS, rack, calibration
12	Air Conditioning	Refrigerant circuit, compressor, HVAC
13	Emissions	Catalyst, EGR, DPF, EVAP, smoke
14	Safety / SRS	Airbag, sensors, belts
15	Vehicle Network	CAN/LIN, gateway, communication faults
16	Lighting / Electronics	Lighting and electronic comfort systems
17	Body / Comfort	Body electrical and comfort
18	ADAS	Radar/camera systems, calibration, TPMS
19	Hybrid / EV	HV battery, inverter, EV diagnostics
20	Tires / Alignment	Tires, balancing, alignment
21	Body Repair / Paint	Accident repair, PDR, paint, ceramic

8.2 Diagnostic Subservices **[CONFIRMED intent]**

Diagnostic → OBD Scan, DTC Reading, DTC Analysis, Live Data, Freeze Frame, Actuator Test, ECU Communication, ABS Diagnosis, Airbag Diagnosis, ESP Diagnosis, Transmission Diagnosis, BCM Diagnosis, BSI Diagnosis, UCH Diagnosis, CAN Bus Diagnosis

8.3 ECU Programming Subservices **[CONFIRMED intent]**

ECU Programming → ECU Flashing, ECU Coding, ECU Adaptation, ECU Reset, Module Programming, Immobilizer Programming, Key Coding, CAN/LIN Configuration, Software Update

[CONFIRMED rule] Only expose services that are technically appropriate for the specific vehicle/module.

8.4 Remaining Category Subservices [CONFIRMED intent]

Category	Subservices
Engine Repair	Engine Diagnosis, Compression Test, Misfire Diagnosis, Cylinder Head, Head Gasket, Valve Repair, Valve Adjustment, Timing Belt, Timing Chain, Piston, Crankshaft, Oil Consumption, Overheating, Turbocharger, Intake System, Throttle Body, Injector, Engine Overhaul
Gearbox	Manual, Automatic, CVT, DCT, AMT, Clutch, Gearbox Oil, Torque Converter, Solenoid, Mechatronics, Shift Problems, Gearbox Diagnosis, Gearbox Programming, Gearbox Overhaul
Electrical	Battery, Alternator, Starter, Wiring, Harness, Fuse Box, Relay, Sensors, Actuators, Lighting, Power Windows, Central Locking, Electronic Modules
Fuel System	Fuel Pump, Injector, Fuel Rail, Fuel Pressure, Throttle Body, MAF, MAP, Oxygen Sensor, EVAP, Fuel Consumption Diagnosis
Cooling	Radiator, Water Pump, Thermostat, Cooling Fan, Expansion Tank, Coolant, Hoses, Bleeding, Overheating Diagnosis
Brakes	Brake Pads, Brake Disc, Caliper, Brake Fluid, Brake Bleeding, ABS, EBD, ESP/ESC, Brake Vibration
Suspension	Shock Absorber, Spring, Control Arm, Ball Joint, Bushings, Stabilizer, Knuckle, Suspension Noise
Steering	Hydraulic Steering, EPS, Steering Pump, Steering Rack, Steering Column, Steering Angle Sensor, Calibration
Air Conditioning	AC Diagnosis, Refrigerant, Leak Detection, Compressor, Condenser, Evaporator, Expansion Valve, Blower, HVAC, Climate Control
Emissions	Exhaust, Catalytic Converter, Oxygen Sensor, EGR, EVAP, DPF, Smoke Diagnosis, Emission Fault Diagnosis
Safety / SRS	Airbag, SRS ECU, Crash Sensor, Seat Belt, Airbag Warning Light, SRS Diagnosis, SRS Reset
Vehicle Network	CAN Bus, LIN Bus, Gateway, Module Communication, Network Faults, Communication Errors
ADAS	Adaptive Cruise Control, Lane Assist, Parking Sensors, Parking Camera, Radar, Camera Calibration, TPMS, Driver Assistance Systems
Hybrid / EV	HV Battery, BMS, Electric Motor, Inverter, On-board Charger, DC/DC, Battery Cooling, HV Safety, EV Diagnostics
Tires / Alignment	Tire Replacement, Tire Repair, Wheel Balancing, Alignment, Tire Wear Diagnosis, Vibration Diagnosis
Body Repair	Accident Repair, Dent Repair, PDR, Paint, Bumper Repair, Polishing, Ceramic, Body Inspection

[CONFIRMED rules] ADAS must only be assigned to vehicles that actually support the relevant system. Hybrid/EV services must only be applied to compatible vehicles.

9. Vehicle × Service Matrix

9.1 Principle [CONFIRMED]

The system must use a **controlled** Vehicle × Service relationship. The canonical source is the database (`vehicle_model_service` – Section 5). The matrix must **NOT** be generated purely by assumptions.

9.2 Illustrative Example – Peugeot 206 [illustrative; each actual assignment must be verified]

Service	Assignment
Diagnostic	YES
Engine Repair	YES
Gearbox	YES
Electrical	YES
Fuel System	YES
Cooling	YES
Brakes	YES
Suspension	YES
Air Conditioning	YES
Hybrid / EV	NO
ADAS	Only if actually supported by the specific vehicle

This table illustrates the **mechanism** of the controlled matrix, not verified per-vehicle truth. Actual assignments must be established per vehicle through the master data workflow (Section 20) and stored canonically.

10. Problem Architecture

[FUTURE content model – DB support NEEDS VERIFICATION]

Problem pages should eventually support:

Problem definition · Symptoms · Possible causes · Diagnostic tree · Required tests · DTC codes · Related ECU · Related vehicle · Related service · Repair solutions · Parts · Warnings · FAQ · Related articles

Example problem concepts: engine vibration, hard starting, overheating, oil consumption, gearbox slipping, ABS warning, check engine, battery drain, AC not cooling.

11. Symptom Architecture

[CONFIRMED principle] Symptoms are **reusable concepts**. A symptom may relate to multiple vehicles, multiple problems, multiple services, multiple DTCs, and multiple articles. Do not duplicate the same symptom unnecessarily.

Example symptoms: rough idle, hard starting, no start, engine vibration, loss of power, high fuel consumption, overheating, abnormal gearbox shifting, brake vibration, steering vibration, warning light.

12. DTC Architecture

[FUTURE connectivity model]

DTC pages/data should eventually connect: DTC Code → ECU / Module → Vehicle → Symptoms → Possible Causes → Diagnostic Tests → Repair Solutions → Related Services → Related Articles.

Example codes: P0300, P0171, P0420, plus ABS-related, SRS-related, and transmission-related codes.

[DO NOT INSERT] Do not invent vehicle-specific DTC compatibility. Only assign DTCs to vehicles when technically verified. A DTC is **not** universal across all vehicles.

13. ECU / Module Architecture

[FUTURE model]

The system should eventually support these module types: Engine ECU, Transmission ECU, ABS, ESP/ESC, Airbag/SRS, BCM, BSI, UCH, Gateway, Immobilizer, Instrument Cluster, Body Control Modules, HVAC module, ADAS modules, CAN/LIN related modules.

Each module may relate to: vehicles, DTCs, diagnostic procedures, repair solutions, programming services, articles.

[CONFIRMED data-quality rule] Never assume the ECU is the same across all variants. Never assume DTC = universal, engine = same across model years, or gearbox = same across variants.

14. Content Strategy & Anti-Thin-Content Rules

14.1 Relationship-First Content **[CONFIRMED intent]**

Content must connect: Vehicle → System → Component → Problem → Symptom → DTC → Service → Repair Solution → Article → FAQ.

Example relationship chain (illustrative – verify per vehicle before publishing): Peugeot 206 → TU3 engine → misfire → P0300/P0301/P0302/P0303/P0304 → ignition/fuel/compression diagnosis → related service → repair solution → technical article.

This relationship structure is **more important** than simply generating many pages.

14.2 Anti-Thin-Content Rule **[CONFIRMED – hard rule]**

Do **NOT** create thousands of nearly identical pages simply by replacing the vehicle name. A page must provide genuine value. Useful content may include:

- technical specifications
- known problems
- symptoms
- diagnostic procedures
- real repair experience
- tools
- measurements
- DTCs
- components
- maintenance intervals
- related services
- FAQs
- diagrams/images where appropriate
- internal links

If a page cannot provide sufficient useful content, do not automatically publish it as indexable content.

14.3 Internal Linking [CONFIRMED intent]

The website should behave like a specialized automotive Wikipedia. Vehicle page → engine page → common problem → DTC → diagnostic service → repair solution → article. Articles should link back to vehicle, service, problem, DTC, part, system. Use contextual internal links; avoid random or excessive links.

14.4 Vehicle (Model) Hub Page – Target Sections [PROPOSED]

Every vehicle/model page should eventually support:

1. Vehicle introduction
2. History
3. Market information
4. Technical specifications
5. Engine options
6. Transmission options
7. ECU information
8. Control modules
9. Electrical architecture
10. Common problems
11. Common symptoms
12. DTC codes
13. Diagnostic information
14. Maintenance schedule
15. Maintenance tasks
16. Available repair services
17. Related parts
18. Repair solutions
19. Related articles
20. FAQ
21. Breadcrumbs
22. Internal links

The page must be useful even before every advanced section is populated. **[CONFIRMED rule]** Do not create empty sections simply to make a page appear complete.

14.5 Vehicle × Service Page – Target Sections [PROPOSED]

Example: </vehicles/peugeot/peugeot-206/diagnostic>. The page should eventually contain:

1. Specialized title
2. Technical introduction
3. Why this vehicle may require this service
4. Symptoms
5. Common causes
6. Diagnostic procedure
7. Required tools
8. ECU/module information
9. Related DTC codes
10. Related parts
11. Repair solutions
12. Approximate cost
13. Mechanic experience / practical notes
14. Safety warnings
15. FAQ
16. Related articles
17. Related vehicle services
18. Breadcrumbs
19. Schema data where appropriate

[DO NOT INSERT] Do not generate fake prices, reviews, or ratings — including inside "approximate cost" sections.

15. URL & Routing Architecture

15.1 Preferred Canonical Structure [PROPOSED target — map against live routes in Phase F]

Pattern	Page
/vehicles	Vehicle index
/vehicles/{brand}	Brand hub
/vehicles/{brand}/{model}	Model hub

Pattern	Page
<code>/vehicles/{brand}/{model}/{service}</code>	Vehicle × service page
<code>/vehicles/{brand}/{model}/{problem}</code>	Vehicle × problem page

Examples: `/vehicles/peugeot` · `/vehicles/peugeot/peugeot-206` · `/vehicles/peugeot/peugeot-206/diagnostic` · `/vehicles/peugeot/peugeot-206/engine-repair` · `/vehicles/peugeot/peugeot-206/gearbox`

The UI may visually show the model page as a hub (Model → Services: Diagnostic, Engine Repair, Gearbox, Electrical), but the canonical SEO URL does **not** need a literal `/services/` segment. Preferred: `/vehicles/peugeot/peugeot-206/diagnostic` — not `/vehicles/peugeot/peugeot-206/services/diagnostic`. The final URL system must be **consistent and stable**.

15.2 Routing Validation Rules [CONFIRMED — hard rule]

Routes must validate relationships. Example: `/vehicles/peugeot/peugeot-206/diagnostic` must resolve **only if**:

1. Peugeot exists,
2. Peugeot 206 exists,
3. Diagnostic service exists,
4. the vehicle-service relationship exists (canonically in `vehicle_model_service`).

If the relationship does not exist → return a proper **404**. Do **NOT** show a fake service page simply because the service slug exists. This is critical for SEO and data integrity.

16. SEO Architecture & Structured Data

16.1 Per-Page Requirements [CONFIRMED]

Every indexable page must have: unique title, unique meta description, canonical URL, correct H1, logical H2/H3 hierarchy, breadcrumbs, internal links, appropriate schema, clean URL, and useful content.

16.2 Technical SEO [CONFIRMED]

robots.txt · sitemap.xml · canonical · index/noindex strategy · 404 handling · redirect strategy · duplicate URL prevention · pagination strategy if required · mobile usability · Core Web Vitals consideration · Search Console readiness.

16.3 Structured Data (Schema) [CONFIRMED]

Use schema appropriate to the actual page type: [Article](#) , [FAQPage](#) (only where genuinely applicable), [BreadcrumbList](#) , [Organization](#) , [WebSite](#) , [LocalBusiness](#) where appropriate.

[DO NOT INSERT] Do not generate fake reviews, fake ratings, fake prices, or fake structured data.

16.4 Images [FUTURE]

Images should eventually support: vehicle images, engine images, component images, diagnostic diagrams, repair illustrations, technical diagrams. Alt text must describe the actual image – no keyword stuffing. AI-generated images, if later introduced, must be accurate enough for their intended technical use and must not be presented as real photographs when they are not.

17. Production, Environment & Security Rules

17.1 Production Safety – Hard Prohibitions [CONFIRMED]

- DO NOT drop/truncate tables or delete production data.
- DO NOT perform destructive migrations.
- DO NOT modify production database schema without approval.
- DO NOT modify production configuration without approval.
- DO NOT expose credentials; DO NOT commit secrets.
- DO NOT hardcode production URLs ([originalauto.ir](#)) throughout the application.
- DO NOT hardcode localhost URLs throughout the application.
- DO NOT create duplicate architecture or a second competing vehicle-service relationship system.
- DO NOT remove existing working functionality without analysis.
- DO NOT restart Apache unless explicitly requested.
- DO NOT restart MySQL unless explicitly requested.

17.2 Change Discipline [CONFIRMED]

Every major change must be: **minimal** · **reversible** · **documented** · **testable**.

17.3 Environment Configuration [CONFIRMED requirement]

The application must support **local development** and **production** (<https://originalauto.ir>) without manually editing PHP files when moving between environments. Configuration must be centralized, with environment detection. Verify the current mechanism in Phase A; centralize if hardcoded values are found [PROPOSED fix – report first].

18. Persian / UTF-8 Requirements

[CONFIRMED] The site is Persian-language. The application must preserve:

- UTF-8 / `utf8mb4` – never introduce `latin1`
 - Persian database content
 - Persian URL handling where appropriate
 - RTL frontend
 - Persian SEO metadata
-

19. Generated Data Policy

[CONFIRMED concepts] Existing generated-data concepts include `app/Data/generated_models` and `app/Data/generated_matrices` .

Rules:

1. Generated data can enrich the application.
 2. Generated data is **NOT** the canonical database relationship.
 3. The AI developer must determine whether generated files are still required after auditing the current architecture.
 4. Do **not** delete generated data merely because it appears redundant.
-

20. Data Quality, Master Data Workflow & Data Status

20.1 Never-Assume Rules [CONFIRMED – hard rules]

Never assume: brand = manufacturer · model = trim · model = generation · service = capability · DTC = universal across all vehicles · ECU = same across all variants · engine = same across all model years · gearbox = same across all variants.

Always account for: model year, generation, engine, transmission, variant, market, and ECU/module differences.

20.2 Master Data Workflow [CONFIRMED recommended flow]

1. Collect candidate brands → 2. Verify brands → 3. Separate market brands from manufacturers → 4. Collect models → 5. Verify models → 6. Identify generations → 7. Identify variants → 8. Identify engines → 9. Identify transmissions → 10. Identify ECU/modules → 11. Map services → 12. Map common problems → 13. Map symptoms → 14. Map DTCs → 15. Map repair solutions → 16. Map parts → 17. Map articles → 18. Generate internal links → 19. Generate SEO metadata → 20. Validate before publication.

20.3 Data Status Model [PROPOSED]

Every imported vehicle record should ideally carry one of:

Status	Meaning
verified	Confirmed against authoritative sources
needs-review	Imported, awaiting verification
historical	Discontinued/historical vehicle
limited-import	Limited import into Iran
inactive	Deliberately hidden
draft	Not ready

[CONFIRMED rule] Do not expose unverified data as authoritative production content.

21. Future Capabilities: Admin, Master Excel & AI Content

21.1 Admin / Data Management [FUTURE]

Future admin capabilities may include: brand, model, variant, engine, transmission, and service management; vehicle-service relationship management; problem, DTC, article, and FAQ management; content review; SEO metadata management; import/export; bulk Excel import. Implementation must respect the current project architecture.

21.2 Excel Master Catalog [FUTURE]

A future master Excel dataset should ideally contain: Brand, Market Brand, Manufacturer, Model, Generation, Variant, Year From, Year To, Engine, Engine Code, Fuel Type, Transmission, Transmission Type, ECU, Modules, Services, Common Problems, Symptoms, DTC, Status, Source, Verification Notes.

[CONFIRMED rule] The Excel file is a **data preparation source**, not automatically a production database dump.

21.3 AI Content System [FUTURE]

The long-term project may include AI-assisted generation of: article drafts, vehicle descriptions, problem pages, DTC explanations, FAQ drafts, image prompts, technical diagrams.

[CONFIRMED rule] AI-generated content must be reviewed. Do not mass-publish unverified automotive technical information.

22. Database Mapping Protocol

[CONFIRMED requirement] Before any migration, the AI developer must produce this mapping – before changing schema, never after:

Current table	Current columns	Current relationships	Target concept	Required change	Risk	Migration required?	Rollback strategy
(from Phase B)	(from Phase B)	(from Phase B)	(from this spec)	(minimal)	(rated)	(yes/no)	(mandatory)

Every row must include a rollback strategy. Any schema change on production requires prior approval.

23. Testing Plan

[CONFIRMED] Every implementation phase must include tests. Test categories:

Category	Coverage
Route tests	All defined and target routes resolve correctly
Vehicle lookup tests	Brand lookup, model lookup
Service lookup tests	Service resolution
Vehicle-service relationship validation	Canonical relationship honored
404 tests	Invalid combinations return proper 404
SEO tests	Titles, descriptions, canonicals
Canonical tests	Correct canonical per page type
Sitemap tests	Valid, complete, no hidden sections
Database integrity	Relationships, constraints, orphans
Persian encoding	UTF-8/utf8mb4 round-trip intact
Mobile layout	Usable on mobile viewports
Security	No credential exposure, no injection regressions
Regression testing	Existing working routes remain functional

24. Acceptance Criteria

[CONFIRMED] The architecture is acceptable when:

1. Vehicle hierarchy is consistent.
2. Brand/model/service relationships use canonical DB data.
3. Invalid vehicle-service combinations return 404.
4. URLs are stable.
5. Canonicals are correct.
6. No duplicate route architecture exists.
7. Persian encoding remains correct.
8. Existing working routes remain functional.

9. SEO metadata is controlled centrally.
10. Production configuration remains safe.
11. No destructive database changes occur without approval.
12. Content architecture supports future scaling.
13. Internal linking can connect vehicles, services, problems, DTCs and articles.
14. The system can expand to hundreds of brands/models/services without architectural duplication.

25. Gap Report Template

[CONFIRMED requirement] Before implementation, produce this table filled from audit findings (Phases A–I):

Area	Current State	Desired State	Gap	Files/Tables	Risk	Recommendation
Architecture	—	—	—	—	—	—
Routing	—	—	—	—	—	—
Database	—	—	—	—	—	—
Vehicles	—	—	—	—	—	—
Services	—	—	—	—	—	—
Vehicle-Service Matrix	—	—	—	—	—	—
DTC	—	—	—	—	—	—
Problems	—	—	—	—	—	—
Symptoms	—	—	—	—	—	—
Articles	—	—	—	—	—	—
SEO	—	—	—	—	—	—
Content	—	—	—	—	—	—
Security	—	—	—	—	—	—
Production	—	—	—	—	—	—
Frontend	—	—	—	—	—	—
Mobile	—	—	—	—	—	—

26. Final Report Template

[CONFIRMED requirement] After implementation, provide:

1. Files changed
 2. Database changes
 3. Routes changed
 4. New features
 5. Removed features
 6. Tests executed
 7. Tests passed
 8. Tests failed
 9. Remaining risks
 10. Rollback instructions
 11. Production deployment instructions
-

27. Final Developer Checklist

- ☐ Read this document completely before opening the code.
- ☐ Completed Phases A–I before any modification; Gap Report reviewed.
- ☐ Verified the real schema before writing any SQL.
- ☐ No DROP/TRUNCATE/destructive migration; no production schema/config change without approval.
- ☐ No credentials exposed; no secrets committed.
- ☐ No hardcoded production or localhost URLs; configuration centralized with environment detection.
- ☐ `vehicle_model_service` remains the single canonical vehicle-service relationship; no competing matrix table created.
- ☐ Generated data enriched where useful, never treated as canonical, never deleted without justification.
- ☐ Invalid brand/model/service/problem combinations return proper 404; no fake pages.
- ☐ URL system consistent, stable, and mapped against live routes.
- ☐ Market brand vs. manufacturer vs. assembler distinguished; collapsed concepts reported.

- ☐ No DTC, ADAS, Hybrid/EV, or vehicle-service assignment inserted without verified technical source.
 - ☐ No fake reviews, ratings, prices, or schema data.
 - ☐ Anti-thin-content rule respected; insufficient pages not published as indexable.
 - ☐ UTF-8/utf8mb4 and RTL preserved; no latin1 introduced.
 - ☐ Hidden sections (Store, Repair Shop Dashboard, User Account) absent from navigation and handled for SEO.
 - ☐ Apache/MySQL not restarted unless explicitly requested.
 - ☐ Every change minimal, reversible, documented, testable.
 - ☐ Test suite (Section 23) executed; results in the Final Report (Section 26).
 - ☐ Final Report delivered, including rollback and deployment instructions.
-

End of Specification. This document is an architectural specification, not permission to modify the project. Never make large architectural assumptions. Never destroy existing data. Never create duplicate systems. Never invent automotive technical information. Never publish unverified vehicle/service/DTC relationships as authoritative.